EFC vs Stage-III Cosmology State Assessment

Structural Coherence Evaluation and Validation Update for Energy-Flow Cosmology in the Post-Tension Weak-Lensing Landscape

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Supersedes: Prior S_8-tension narrative in EFC validation documents

ABSTRACT

This document provides a Stage-III cosmology update to the Energy-Flow Cosmology (EFC) validation program. Recent weak-lensing recalibrations — KiDS Legacy (Wright et al., A&A 703, A158, 2025) and HSC Y3 with DESI clustering-redshift calibration (Choppin de Janvry et al., arXiv:2511.18134, 2025) — shift the S_8 landscape upward, reducing earlier 2–3σ tension with CMB-inferred growth to sub-1σ levels. We reassess EFC's observational position under these updates, reframing its role from "tension resolution" to "precision structural competitor." We introduce a Structural Coherence Evaluation (SCE) protocol that distinguishes observation-native quantities (O-layer) from model-dependent summary statistics (M-layer) and define falsifiable mapping-level tests for the EFC lensing response Σ(k,z). The background coupling $\beta = 0.08$, locked by BAO distance measurements, yields $\sigma_8 \approx 0.805$ — consistent with the converging Stage-III central values without post-hoc adjustment.

Keywords: *Energy-Flow Cosmology · S_8 tension · weak lensing · Stage-III surveys · KiDS Legacy · HSC Y3 · DESI clustering-redshift · modified gravity · structural coherence evaluation · cosmological model comparison · observation-native ontology · lensing response*

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Part 1 — Validation Ledger Entry

VL-2026-02-10: S_8 Landscape Convergence and EFC Repositioning

Trigger: Stage-III final updates move the weak-lensing S_8 baseline upward, changing the role S_8 plays in EFC model positioning.

Convention: Throughout, $S_8 \equiv \sigma_8\sqrt{(\Omega_m/0.3)}$. We treat reported "combined" constraints according to the methodology of the cited analysis, not as an official cross-consortium consensus product.

Observational update (Stage III):

Probe / analysis	Updated S8	Notes	Reference
KiDS Legacy (cosmic shear)	0.815 (+0.016/–0.021)	0.73σ consistent with Planck	Wright et al. (2025) [1]
HSC Y3 (DESI-calibrated n(z))	0.805 ± 0.018	Upward shift vs earlier HSC Y3	Choppin de Janvry et al. (2025) [2]
Combined WL (KiDS+DES+HSC)	0.813 (+0.009/–0.010)	Combined within arXiv:2511.18134	Choppin de Janvry et al. (2025) [2]
CMB lensing (combined)	0.828 ± 0.012	Reference point for growth	Choppin de Janvry et al. (2025) [2]

Note: Planck PR4 [4] shifts S_8 closer to LSS and reduces the S_8 tension. We do not hardcode a PR4 σ_8 value here pending verification of the exact table entry.

Cause of shift: Improved photo-z calibration and validation (including DESI-based clustering-redshifts for HSC), improved image simulations, and expanded calibration datasets — primarily attributed to improved control of calibration systematics rather than requiring new physics.

Residual tension:

- KiDS Legacy alone: negligible vs Planck (0.73σ).
- Some analyses still report residual small-scale sensitivity and modeling dependence, e.g. baryonic/small-scale handling in HSC Y3 contexts [3].
- DES Y3 not yet recalibrated with DESI clustering-z — earlier values (~0.76–0.77) remain current for that survey.

EFC impact assessment:

EFC channel	Updated role	Status
β background (σ8 ≈ 0.805)	Matches converging Stage-III amplitude	Strengthened
Σ(k,z) lensing response	Precision diagnostic for scale-dependent response	Role shifted
T(z) regime consistency	GR recovery at high z	Unchanged

Narrative update:

X **DEPRECATED:** "EFC resolves a 3σ S_8 tension."

> **CURRENT:** "EFC's background coupling yields an amplitude ($\sigma_8 \approx 0.805$) consistent with the converging Stage-III lensing landscape; Σ(k,z) becomes a model-selection diagnostic for scale-dependent matter–lensing

response.”

Action items:

1. Re-run $\Sigma(k,z)$ fits against KiDS-Legacy data ($\Sigma = 1$ locked vs Σ free). Compare $\Delta\chi^2$, AIC, BIC.
2. If Σ improvement persists → genuine explanatory channel (small-scale structure).
3. If Σ improvement collapses → Σ was absorbing pre-final calibration systematics.
4. Update all paper introductions to reflect Stage-III convergence.
5. Falsification criteria F5 (CMB lensing $\sigma_8 > 3\sigma$ from EFC): margin now tighter but still not triggered (-0.71σ).
6. Add Stage-III update footnote to all existing/forthcoming papers (see Part 2).

Part 2 — Paper Section Draft

“EFC in the Post-Tension Landscape”

For use in paper introduction or discussion section.

Recent Stage-III updates have shifted late-time structure-growth constraints upward relative to early weak-lensing results. The final KiDS Legacy cosmic-shear analysis reports S_8 = 0.815 (+0.016/–0.021), consistent with Planck at the 0.73σ level [1]. A reanalysis of HSC Year 3 with DESI-based clustering-redshift calibration yields S_8 = 0.805 ± 0.018, and the same framework reports a combined Stage-III weak-lensing constraint of S_8 = 0.813 (+0.009/–0.010) [2]. These shifts are widely attributed to improved photometric-redshift calibration and validation rather than a requirement for new physics.

This convergence does not weaken EFC’s background channel; it tightens the claim. With a single-parameter background coupling (β = 0.08), EFC yields $\sigma_8 \approx 0.805$, coinciding with the recalibrated HSC Y3 central value and lying within ~0.5σ of the combined weak-lensing constraint. β was locked by BAO distance measurements (BOSS DR12) and then propagated to growth; the σ_8 prediction is a consequence of the same background modification, not a separate fit. The scientific emphasis therefore shifts from “tension resolution” to “structural prediction”: EFC does not require a large mismatch to motivate itself, but instead aims to match first-order growth amplitude while producing second-order, structured deviations that remain testable.

In this setting, the lensing-response channel Σ(k,z) moves from crisis intervention to precision diagnostics. With bulk S_8 largely stabilized, the relevant question is whether weak-lensing data prefer additional scale-dependent freedom in the matter–lensing response, particularly on small scales where modeling sensitivity can persist [3]. This becomes a model-selection problem (locked vs free Σ), rather than a narrative about a single global “S_8 tension.”

Accordingly, the discriminating tests shift toward structured signatures: scale-dependent growth, environment-dependent clustering, lensing response versus dynamical mass, and cluster shear profiles — domains where EFC’s constitutive coupling predicts deviations from ΛCDM beyond first-order amplitude matching.

Stage-III Update Footnote

For insertion in existing/forthcoming papers:

> *Since submission, the Stage-III weak-lensing landscape has shifted upward: KiDS Legacy reports S8 = 0.815 (+0.016/–0.021) (Wright et al. 2025), and HSC Y3 recalibrated with DESI clustering-redshifts yields S8 = 0.805 ± 0.018 (Choppin de Janvry et al. 2025). The EFC background-coupling prediction (σ8 ≈ 0.805) is consistent with these updated values. The S8 tension motivation discussed in this work should be read in light of this convergence; the structural predictions and falsification criteria remain unchanged.*

Part 3 — Updated EFC Positioning Summary

Category transition:

From: “Alternative model that explains a tension”

To: “Alternative model that matches standard cosmology at first order and predicts second-order structural deviations”

Strengths in new landscape:

- $\beta = 0.08 \rightarrow \sigma_8 = 0.805$ matches converging data without tuning
- Growth suppression via Hubble friction (correct mechanism, correct direction)
- One transition function T(z) governs all regimes (no patches)
- Built-in predictions for scale-dependent and environment-dependent effects
- Five pre-registered falsification criteria, none triggered

Challenges in new landscape:

- Reduced motivation from S_8 tension (the “problem” is smaller)
- Σ channel needs re-validation against updated baselines
- Must demonstrate second-order predictions are testable with current/upcoming data
- Self-consistent EFC background (not ΛCDM-calibrated) still needed

Key discriminating tests (Stage IV era):

Test	EFC prediction	ΛCDM prediction	Data source
Scale-dependent growth f(k,z)	Entropy-gradient modulation	Scale-independent (linear regime)	Euclid, DESI DR3
Lensing vs dynamical mass	$\Sigma \neq 1$ at low-z, small scales	$\Sigma = 1$ everywhere	Euclid clusters, Rubin
Environment-dep. clustering	Entropy-gradient correlations	No environment dependence	eROSITA + Euclid
ISW cross-corr. amplitude	Cancellation from μ/μ'	Standard ISW	DESI tracers × Planck/ACT
Cluster abundance N(M,z)	Modified growth function	Standard Press-Schechter	eROSITA, SPT-3G

The S_8 convergence makes EFC less of a “problem solver” and more of a “precision model with new testable signatures” — a step up in scientific maturity, not down.

Part 4 — SCE Protocol Section (journal-ready)

Structural Coherence Evaluation Protocol for Weak-Lensing Comparisons

When comparing gravitational theories that modify growth, lensing, or background expansion, the Structural Coherence Evaluation (SCE) framework [5] requires that comparisons respect the processing provenance of derived quantities. We apply this protocol to the weak-lensing S_8 parameter as follows.

Layer classification. SCE classifies quantities into three layers by their dependence on theoretical assumptions:

- **O-layer** (Observation-Native Ontology): Quantities extracted from data with minimal theoretical input. For weak lensing: shear two-point correlation functions $\xi_\pm(\theta)$, convergence power spectra $C_l^{\kappa\kappa}$, and tomographic bin statistics.
- **M-layer** (Mapping Transparency): Quantities produced by passing O-layer data through a model-dependent pipeline. S_8 is an M-layer product: it requires assumptions about the lensing kernel (GR Poisson equation, $\Sigma = 1$), growth law (GR linear perturbation theory), background expansion (Friedmann + ΛCDM), light propagation (constant c), and matter power spectrum shape (ΛCDM transfer function).
- **T-layer** (Theory Consumption): Theories consume M-layer products as compressed constraints. When a theory consumes S_8, it inherits the mapping assumptions embedded in the extraction pipeline.

Consequence for model comparison. Agreement in S_8 between EFC and ΛCDM demonstrates M-coherence: both frameworks produce compatible values for a GR-compressed summary statistic. This is a necessary consistency check but not a sufficient condition for structural equivalence between theories. Two frameworks may produce identical S_8 while predicting different shear correlation patterns, because S_8 compresses scale-dependent, redshift-dependent, and environment-dependent information into a single number.

Observation-level test. The structurally informative comparison operates at the O-layer. For the lensing channel, this means comparing tomographic shear likelihoods under the ΛCDM mapping ($\Sigma = 1$ locked) versus the EFC mapping ($\Sigma(k,z)$ free):

$$C_l^{\kappa_i\kappa_j} = \int d\chi\ [W_i(\chi)\ W_j(\chi)\ /\ \chi^2] \cdot \Sigma^2(k,z) \cdot P_\delta(k,z)$$

where $\Sigma = 1$ recovers standard GR and $\Sigma(k,z) \neq 1$ implements the EFC lensing response.

Falsification criterion. Under SCE's self-imposed falsifiability principle, the additional mapping freedom must be statistically justified at the O-layer. If the extended mapping (Σ free) does not improve the observation-level likelihood beyond information-criteria penalties ($\Delta\chi^2$ insufficient to overcome ΔAIC > 0, ΔBIC > 0 from extra parameters), the additional degrees of freedom are disfavored and the simpler mapping is preferred. This ensures that structural motivation does not substitute for empirical evidence.

Application to current data. EFC's background coupling ($\beta = 0.08$) yields $\sigma_8 \approx 0.805$, demonstrating M-coherence with the converging Stage-III lensing landscape (KiDS Legacy: 0.815, HSC Y3+DESI: 0.805, combined WL: 0.813). The pending O-layer test — refitting KiDS Legacy with Σ free versus $\Sigma = 1$ — will determine whether the data prefer additional scale-dependent response freedom once the bulk amplitude mismatch is resolved by improved calibration.

Part 5 — Figure Caption

Figure X. SCE layer classification of the weak-lensing S_8 parameter.

The Structural Coherence Evaluation (SCE) framework classifies quantities by their processing provenance — the chain of theoretical assumptions required to derive them from raw data. **O-layer** (top, green): observation-native quantities requiring no gravitational theory — shear two-point correlations $\xi_\pm(\theta)$, convergence power spectra $C_l^{\kappa\kappa}$, and tomographic bin statistics. **M-layer** (center, amber): model-dependent mapping pipeline that converts O-layer data into derived parameters. $S_8 \equiv \sigma_8\sqrt{(\Omega_m/0.3)}$ is produced here under assumptions of GR lensing ($\Sigma = 1$), GR linear growth, Friedmann background, constant c, and ΛCDM-parametrized P(k,z). Blue labels indicate where EFC modifies the mapping: $\Sigma(k,z) \neq 1$ in the lensing kernel, modified Hubble friction H_{EFC} in the growth law, and $P_{eff} = P \cdot \Sigma^2$ in the power spectrum. Grey labels show the ΛCDM assumptions each mapping step requires. **T-layer** (bottom, red): theories consume the compressed M-layer output as fit targets. Agreement in S_8 between frameworks demonstrates M-coherence (mapping-level consistency) but not structural equivalence; the observation-level (O-coherence) test compares tomographic shear likelihoods under $\Sigma = 1$ versus $\Sigma(k,z)$ free (purple box). Under SCE's self-imposed falsifiability principle, extra mapping freedom must be justified by information-criteria improvement (ΔAIC, ΔBIC) at the O-layer — structural motivation alone is insufficient.

Relationship to Existing EFC Papers

This document does not replace earlier EFC analyses. The four-channel consistency test (BAO + RSD + CMB lensing + Lyα P1D) and the regime-activated lensing response paper [6] remain valid within their stated observational baselines. The present document updates the interpretation context in which those results should be read: the S_8 tension that partially motivated the lensing response channel has been reduced by improved calibration, while the background coupling prediction ($\sigma_8 \approx 0.805$) has been independently confirmed by converging Stage-III measurements. The SCE protocol introduced here (Part 4) provides a formal methodological layer that applies retroactively to all EFC–ΛCDM comparisons involving derived parameters. Future EFC papers should reference this document for the current observational baseline and for the SCE layer classification of weak-lensing quantities.